

Exp no:5**Integration with External Systems using Hyperledger Fabric****Aim:**

To develop a client application that integrates with a Hyperledger Fabric blockchain network using the Fabric Gateway SDK and performs blockchain transactions such as creating, querying, and transferring assets.

What this Experiment demonstrates

None

Fabric Network



JavaScript Chaincode



Fabric Gateway SDK



Node.js Client Application



Submit Transaction



Query Ledger



Create Asset



Transfer Asset



Display Result

procedure:**1. Start the Fabric network**

None

```
cd ~/fabric-dev/fabric-samples/test-network
```

```
./network.sh up createChannel -ca
```

```
using orderers
. scripts/orderer.sh mychannel
['0' -eq 1]
res=0
omitted chaincode definition for chaincode 'basic' on channel 'mychannel':
ersion: 1.0, Sequence: 2, Endorsement Plugin: escc, Validation Plugin: vsc, Approvals: [Org1MSP: true, Org2MSP: true]
tatus: 405

{"error": "cannot join: channel already exists"}

channel 'mychannel' created
oining org1 peer to the channel...
sing organization 1
peer channel join -b ./channel-artifacts/mychannel.block
res=1
peer channel join -b ./channel-artifacts/mychannel.block
res=1
peer channel join -b ./channel-artifacts/mychannel.block
res=1
peer channel join -b ./channel-artifacts/mychannel.block
res=1
026-09-08 17:46:56.137 IST 0001 INFO [channelCmd] InitCmdFactory → Endorser and orderer connections initialized
rror: proposal failed (err: bad proposal response 500: cannot create ledger from genesis block: ledger [mychannel] already exists with state [ACTIVE])
fter 5 attempts, peer0.org1 has failed to join channel 'mychannel'
```

2. Deploy the JavaScript chaincode

None

```
./network.sh deployCC \
-ccl javascript \
-ccn basic \
-ccp ../asset-transfer-basic/chaincode-javascript
```

3. Set the Fabric environment

None

```
export PATH=${PWD}/../bin:$PATH
export FABRIC_CFG_PATH=$PWD/../config/

export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
```

```
export
CORE_PEER_TLS_ROOTCERT_FILE=${PWD}/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt

export
CORE_PEER_MSPCONFIGPATH=${PWD}/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp

export CORE_PEER_ADDRESS=localhost:7051

export
ORDERER_CA=${PWD}/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem
```

4. Go to the Gateway application

```
None
cd ../asset-transfer-basic/application-gateway-javascript
```

5. Install Node.js dependencies

```
None
npm install
```

6. Run the client application

```
None
node app.js
```

The application should connect to Fabric through the **Fabric Gateway SDK** and perform the operations defined by the sample application.

7. If your application uses **npm start**

Check `package.json`:

None

```
cat package.json
```

If it contains a start script, you can also use:

None

```
npm start
```

Otherwise use:

None

```
node app.js
```

```

src/client.js:162:26:
    at Object.onReceiveStatus (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client.js:193:36)
    at Object.onReceiveStatus (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client-interceptors.js:367:141)
    ... 2 lines matching cause stack trace ...
    at process.processTicksAndRejections (node:internal/process/task_queues:85:11) {
  code: 10,
  details: [
    {
      address: 'peer0.org1.example.com:7051',
      message: 'chaincode response 500, The asset asset70 does not exist',
      mspId: 'Org1MSP'
    }
  ],
  cause: Error: 10 ABORTED: failed to endorse transaction, see attached details for more info
    at callErrorFromStatus (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/call.js:32:19)
    at Object.onReceiveStatus (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client.js:193:76)
    at Object.onReceiveStatus (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client-interceptors.js:367:141)
    at Object.onReceiveStatus (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client-interceptors.js:327:181)
    at /home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/resolving-call.js:13:78
    at process.processTicksAndRejections (node:internal/process/task_queues:85:11)
    for call at
    at Client.makeUnaryRequest (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@grpc/grpc-js/build/src/client.js:161:32)
    at /home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@hyperledger/fabric-gateway/dist/client.js:3:62
    at new Promise (<anonymous>)
    at GatewayClientImpl.endorse (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@hyperledger/fabric-gateway/dist/client.js:37:16)
    at ProposalImpl.endorse (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@hyperledger/fabric-gateway/dist/proposal.js:43:52)
    at process.processTicksAndRejections (node:internal/process/task_queues:104:5)
    at async ContractImpl.submitAsync (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@hyperledger/fabric-gateway/dist/contract.js:48:29)
    at async ContractImpl.submit (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/node_modules/@hyperledger/fabric-gateway/dist/contract.js:40:27)
    at async updateNonExistentAsset (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/src/app.js:267:9)
    at async main (/home/ram/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript/src/app.js:121:9) {
  code: 10,
  details: 'failed to endorse transaction, see attached details for more info',
  metadata: Metadata {
    internalRepr: [Map],
    opaqueData: Map(0) {},
    options: {}
  }
},
  transactionId: '73e96aeadd6d01d56806607ddd2dad687860f1d0088df031225c31a2b64879e6f'

```

```

--(ram@ram)~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript
$ cd ~/fabric-dev/fabric-samples/test-network

export PATH=${PWD}/../bin:$PATH
export FABRIC_CFG_PATH=${PWD}/../config/

export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export CORE_PEER_TLS_ROOTCERT_FILE=${PWD}/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
export CORE_PEER_MSPCONFIGPATH=${PWD}/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051

peer chaincode query \
  -n mychannel \
  -n basic \
  -c '{"Args":["GetAllAssets"]}'
{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"},{"AppraisedValue":1300,"Color":"yellow","ID":"asset178887063381","Owner":"Saptha","Size":5},{"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"},{"AppraisedValue":1500,"Color":"Green","ID":"asset20","Owner":"Arun","Size":8},{"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"},{"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"},{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"},{"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]

```